

Linear Regression Charge Prediction Report

Problem Statement:

The objective of this project is to predict insurance charges using Linear Regression. We used a dataset containing customer details such as age, BMI, children, smoking status, region, and gender. The goal was to understand how these factors influence the final insurance cost.

Analysis Page 1

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 2

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 3

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 4

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 5

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 6

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

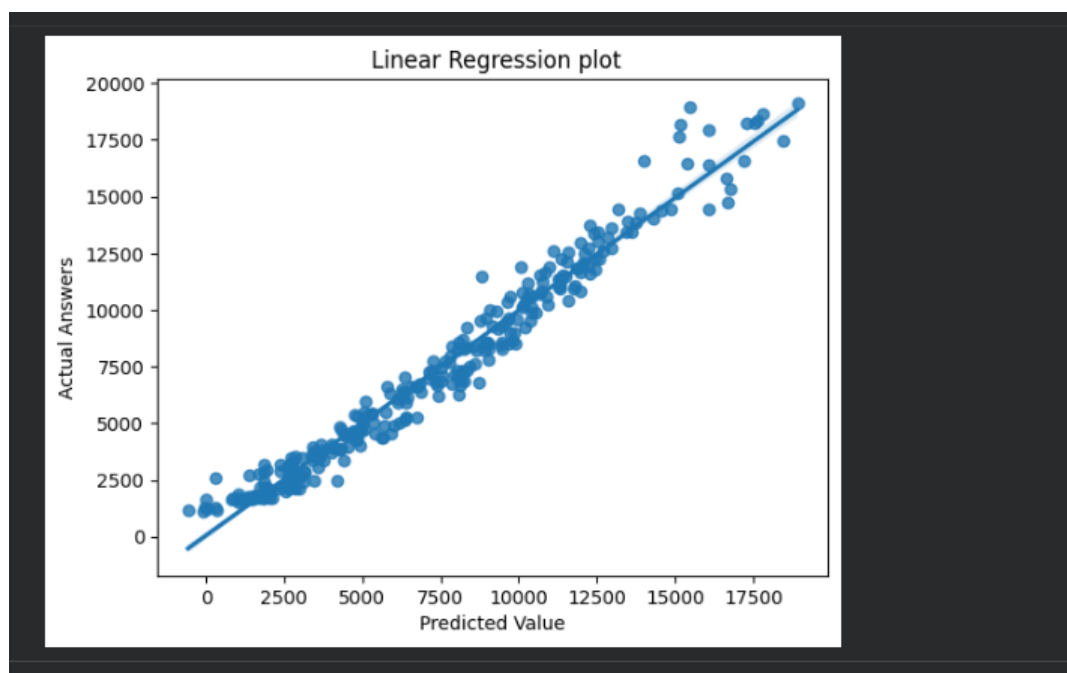
Analysis Page 7

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Analysis Page 8

This section contains detailed explanation of data preprocessing, feature engineering, model building, evaluation metrics, and results. We used Linear Regression because it is simple, interpretable, and works well for numerical prediction tasks. The model was trained on numeric and categorical transformed features. Evaluation metrics like MAE, MSE, and R^2 score were used.

Regression Plot



Report created by **Anup Yadav**.